

Relations & Equivalence

2.AB PreIB Maths – Exam D

Unless specified otherwise, you are to **always** (at least briefly) explain your reasoning. Even in closed questions.

Product of Sets & Relations

Let $A = \{\text{🍌}, \text{🍌🍌}\}$ and $B = \{\text{🍌}, \text{🍇}, \text{🍪}\}$. Check all **correct** statements of the ones below. You **don't have to** explain yourself. [25 %]

- ☐ The set $\{(\text{🍪}, \text{🍌🍌}), (\text{🍪}, \text{🍪}), (\text{🍇}, \text{🍌})\}$ is a relation from B to A .
- ☐ The set $(A \times B) \cap (B \times A)$ is **empty**.
- ☐ There are **more** relations from B to A than there are relations from A to B .
- ☐ The element $(\text{🍌🍌}, \text{🍌🍌})$ lies in $A \times A$.
- ☐ The set $B \times B$ has 9 elements.

A relation R from A to B is called **injective** if whenever $a_1 R b$ and $a_2 R b$, then $a_1 = a_2$ for $a_1, a_2 \in A$ and $b \in B$. In words, no two **different** elements from A can be related to the **same** element in B . Mark only those relations below that are injective. [25 %]

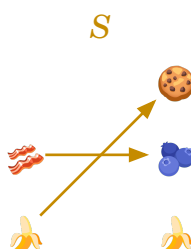
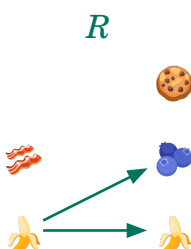


Bonus Problem

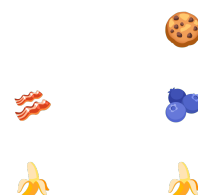
The **union** of relations R and S from A to B is the relation written as $R \cup S$ from A to B that is defined by [10 %]

$$a(R \cup S)b \text{ in the case that } aRb \text{ or } aSb.$$

for any elements $a \in A, b \in B$. Draw the picture for the relation $R \cup S$ if the pictures of R and S are as below.



Draw $R \cup S$ here.



Equivalence

Recall that the relation of *equivalence* is given by three conditions:

[25 %]

- **reflexivity**: every element is equivalent to itself;
- **symmetry**: if a is equivalent to b , then b is equivalent to a ;
- **transitivity**: if a is eq. to b and b is eq. to c , then a is eq. to c .

Determine if the relation

$$R = \{(1, 1), (1, 2), (2, 2), (3, 2), (4, 4), (4, 1)\}$$

is an equivalence on the set $A = \{1, 2, 3, 4, 5\}$. If not, encircle all the pairs below that must be added to R because of one of the rules above.

$(3, 3)$	$(2, 3)$	$(1, 5)$	$(5, 2)$	$(5, 5)$
$(2, 4)$	$(1, 3)$	$(3, 4)$	$(1, 4)$	$(5, 1)$

Describe using a picture an equivalence on the set $A = \{1, 2, 3, 4, 5\}$ which has the following **classes of equivalence**:

[25 %]

- $\{1, 2\}$, $\{3, 4\}$ and $\{5\}$.
- $\{1, 5\}$ and $\{2, 3, 4\}$.

Bonus Problem

Draw a relation on the set $A = \{1, 2, 3, 4, 5\}$, which is

[10 %]

- symmetric and transitive but **not** reflexive.
- reflexive and transitive but **not** symmetric.

(a)		(b)	
5	5	5	5
4	4	4	4
3	3	3	3
2	2	2	2
1	1	1	1